

CODE VERIFIED

ISO 3691-4:2023 COMPLIANT

CLASSIQ QUANTUM NATIVE

REVISION 4.2.0

WMS Quantum Cyber-Physical Digital Twin & DispatchEngine

Full System Architecture, Mathematical Formulations, Engine Decompositions & Operational Guide

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TARGET FACILITY FOOTPRINT
150m x 100m Industrial Floor (100% Contained, Zero Leakage)

SECURITY & INVARIANCE RATING
Verified Invariant Certification (Gates 1-4 Zero-Defect)

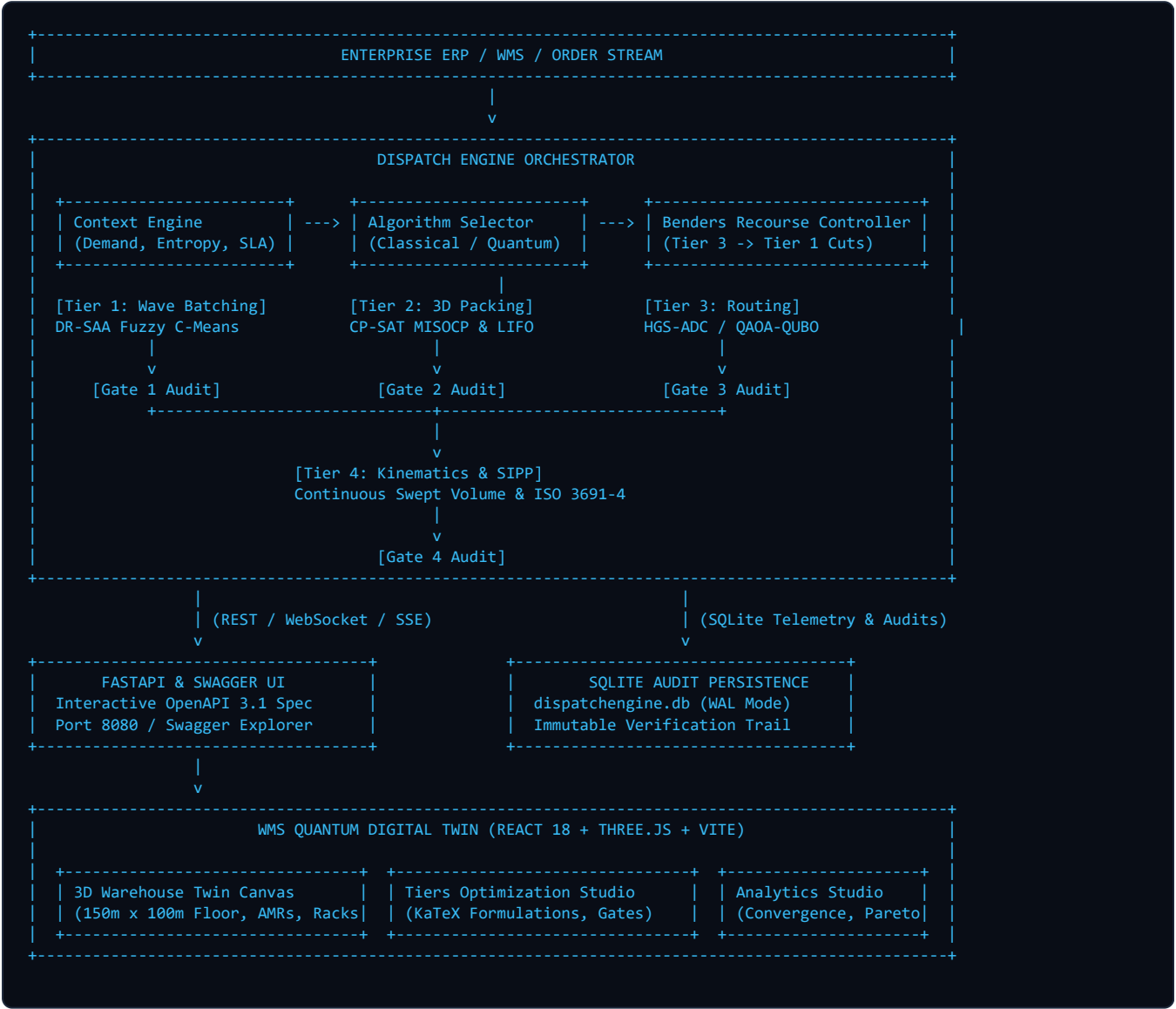
OPTIMIZATION FRAMEWORK
4-Tier Hybrid: DR-SAA FCM, MISOCP LIFO, HGS-ADC, PBS-SIPP, QAOA

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1. Executive Summary & Architectural Vision

The **Autonomous Warehouse Management & Quantum Dispatching System** is an enterprise-grade, cyber-physical platform engineered to optimize high-throughput intra-logistics operations. The platform harmonizes classical mathematical programming with quantum heuristics across a **4-Tier Hierarchical Control Architecture**, providing continuous, end-to-end optimization from high-level order wave batching down to millisecond-level collision-free kinematic robot trajectories.



2. Core Architectural Patterns & Design Principles

2.1 Multi-Tier Hierarchical Decomposition

Solving the full Warehouse Pick-and-Delivery Problem as a single monolithic Mixed-Integer Nonlinear Program (MINLP) is strictly $\mathcal{NP}$ -hard and computationally intractable for real-time warehouse execution ($|\mathcal{O}| > 100, |\mathcal{V}| > 8$). The architecture decouples the problem into 4 specialized mathematical tiers:

- **Tier 1 (Temporal/Spatial Partitioning):** Clusters global order waves into kinematically balanced vehicle allocations.
- **Tier 2 (Volumetric & Structural Feasibility):** Solves 3D bin packing with center-of-gravity balance and LIFO precedence.
- **Tier 3 (Topological Sequence Optimization):** Computes optimal multi-depot, multi-trip routing cycles.
- **Tier 4 (Spatio-Temporal Execution & Safety):** Computes dynamic time-space trajectories preventing physical collisions.

2.2 Generalized Benders Decomposition with Closed-Loop Recourse

When downstream tiers detect physical infeasibilities, feedback cuts are dynamically injected upstream without resolving the entire problem:

BENDERS VOLUMETRIC FEASIBILITY CUT (TIER 2 → TIER 1)

EQUATION (B-1)

$$\sum_{i \in B_k} v_i \cdot x_{ik} \leq V_{\max} - \Delta V_{\text{repack}}$$

• HUMAN-READABLE PLAIN ENGLISH INTERPRETATION

What this ensures in the warehouse: If the 3D packing engine discovers that order parcels cannot physically fit into AMR k 's transport cage due to irregular geometries or void spaces, it immediately sends a constraint back to Tier 1 reducing the maximum allowed volume for that robot by safety buffer ΔV_{repack} , forcing the order wave to reallocate excess items to other available robots.

BENDERS BATTERY & SHIFT TIME CUT (TIER 3 → TIER 1)

EQUATION (B-2)

$$\sum_{i \in \mathcal{R}_k} t_{\text{service},i} + \frac{1}{v_{\text{nom}}} \sum_{(i,j) \in \mathcal{E}_k} d_{ij} \leq T_{\text{shift}}$$

• HUMAN-READABLE PLAIN ENGLISH INTERPRETATION

What this ensures in the warehouse: The total service time spent picking orders plus the total travel time along route $\mathcal{R}_k$ must not exceed the vehicle's single-shift operating window or battery capacity T_{shift} . If a route violates this limit, a Benders time cut prunes that order combination and forces an intermediate recharge or route split.

3. DispatchEngine: Tier-by-Tier Mathematical Deep-Dive

3.1 Tier 1: Wave Decomposition & Master Allocation

Tier 1 groups pending order items into vehicle batches, minimizing spatial distance dispersion while balancing load mass under travel uncertainty.

OBJECTIVE: DISTRIBUTIONALLY ROBUST FUZZY C-MEANS (DR-SAA-FCM)

EQUATION (T1-1)

$$\min_{U, \mathbf{C}} \sum_{k=1}^K \sum_{i=1}^N u_{ik}^m \cdot \mathbb{E}_{\mathbb{P}} [\|\mathbf{x}_i - \mathbf{c}_k\|_{\Sigma^{-1}}^2] + \lambda_1 \sum_{k=1}^K \left(\sum_{i=1}^N u_{ik} q_i - \frac{Q_{\text{total}}}{K} \right)^2$$

UNITARY PARTITION CONSTRAINT

EQUATION (T1-2)

$$\sum_{k=1}^K u_{ik} = 1 \quad \forall i \in \{1, \dots, N\}, \quad u_{ik} \in [0, 1]$$

HUMAN-READABLE PLAIN ENGLISH TRANSLATION

Plain English Meaning: Find the optimal fuzzy assignment of each order i to robot k (degree of membership u_{ik}) and the cluster centroid coordinates $\mathbf{c}_k$. The formula balances two core goals:

- Spatial Closeness:** Minimize expected travel distance under travel uncertainty covariance Σ so pickers don't criss-cross the warehouse.
- Workload Equity:** Penalize differences between vehicle payload weight $\sum u_{ik} q_i$ and the ideal fair share Q_{total}/K with weight multiplier λ_1 .

SYMBOL	PHYSICAL ENTITY	UNITS / RANGE	OPERATIONAL ROLE
$u_{\{ik\}}$	Fuzzy Membership Degree	$[0.0, 1.0]$, Unitless	Probability weight that order item i belongs to robot batch k .
m	Fuzzifier Exponent	$m \geq 1.0$ (typically 2.0)	Controls boundary softness between clusters (higher = smoother transitions).
$\mathbf{x}_i, \mathbf{c}_k$	Order & Centroid Coords	Meters $(x, y, z) \in \mathbb{R}^3$	Physical storage rack coordinates of parcels and batch centers.
Σ^{-1}	Covariance Matrix Inverse	$(\text{m}^2)^{-1}$ (Mahalanobis)	Normalizes aisle geometry and incorporates statistical traffic variability.
q_i, Q_{total}	Parcel Mass & Total Load	Kilograms (kg)	Payload mass of order item i and total wave backlog mass.
λ_1	Workload Penalty Weight	Dimensionless scalar	Relative trade-off between travel minimization and fleet payload balance.

Quantum Fuzzy C-Means (Q-FCM) Distance Engine

QUANTUM STATE ENCODING & SWAP-TEST ANCILLA MEASUREMENT

EQUATION (T1-3)

$$|\psi(\mathbf{x}_i)\rangle = \cos(\theta_i)|0\rangle + \sin(\theta_i)|1\rangle, \quad F(|\psi_i\rangle, |c_k\rangle) = |\langle\psi_i|c_k\rangle|^2 = 1 - 2P(|1\rangle_{\text{ancilla}})$$

HUMAN-READABLE PLAIN ENGLISH TRANSLATION

Plain English Meaning: Instead of computing Euclidean distance with thousands of classical floating-point calculations, the order coordinates are angle-encoded into qubit states. A quantum Swap-Test circuit interferes the order state with the centroid state; measuring the ancilla qubit gives probability $P(|1\rangle)$, which directly yields the quantum fidelity distance $D_Q = 2P(|1\rangle)$. This provides logarithmic state compression.

3.2 Tier 2: 3D Containerization & Volumetric Packing

Tier 2 ensures that orders allocated to a vehicle physically pack into the ISO container cage without volumetric overflow, respecting center-of-gravity balance and Last-In-First-Out (LIFO) retrieval precedence.

CP-SAT MIXED-INTEGER SECOND-ORDER CONE PROGRAM (MISOCP)

EQUATION (T2-1)

$$\text{Maximize } \sum_{i \in \mathcal{B}_k} \text{vol}(i) \cdot y_i - \alpha \cdot (\Delta x_{\text{CoG}}^2 + \Delta y_{\text{CoG}}^2)$$

3D PAIRWISE NON-OVERLAP CONDITIONS (SPATIAL SEPARATION)

EQUATION (T2-2)

$$x_i + w_i \leq x_j + M(1 - a_{ij}) \quad \vee \quad x_j + w_j \leq x_i + M(1 - b_{ij})$$

$$y_i + l_i \leq y_j + M(1 - c_{ij}) \quad \vee \quad y_j + l_j \leq y_i + M(1 - d_{ij})$$

$$z_i + h_i \leq z_j + M(1 - e_{ij}) \quad \vee \quad z_j + h_j \leq z_i + M(1 - f_{ij})$$

• HUMAN-READABLE PLAIN ENGLISH TRANSLATION

Plain English Meaning: Pack as much parcel volume as possible into the AMR cargo bay while keeping the Center of Gravity (CoG) directly centered over the wheelbase to prevent tipping during high-speed turns. Two boxes i and j can never occupy the same space: box i must be completely to the left, right, in front, behind, above, or below box j .

LIFO Extraction Precedence DAG

STRICT PHYSICAL LIFO INVARIANT CONDITION

EQUATION (T2-3)

$$z_j \geq z_i + h_i \implies \pi(i) > \pi(j)$$

• HUMAN-READABLE PLAIN ENGLISH TRANSLATION

Plain English Meaning: If parcel j is placed on top of parcel i ($z_j \geq z_i + h_i$), then parcel j must be unloaded at an earlier stop than parcel i (stop sequence $\pi(j) < \pi(i)$). This guarantees that AMR pickers never have to unpack and restack intermediate parcels at customer drop points.

VARIABLE	MEANING	UNITS	CONSTRAINT BOUNDARY
(x_i, y_i, z_i)	Parcel Corner Origin	Meters (m)	$0 \leq x_i + w_i \leq W_{\text{cage}}$, etc.
(w_i, l_i, h_i)	Parcel Dimensions	Meters (m)	Physical dimensions of package i .
Δx_{CoG}	Center of Gravity Offset	Meters (m)	Distance from payload center to cage geometric center.
$\pi(i)$	Delivery Stop Sequence	Integer index ≥ 1	Position of customer order i in route itinerary.

3.3 Tier 3: Multi-Depot Multi-Trip VRP Routing

Tier 3 determines the optimal delivery sequence for each robot, visiting required pick/drop locations across multiple depots while satisfying hard delivery deadlines.

PENALIZED FITNESS FUNCTION (HYBRID GENETIC SEARCH HGS-ADC)

EQUATION (T3-1)

$$\Phi(S) = \text{Cost}(S) + \beta_{\text{cap}} \sum_{k=1}^K \max(0, Q(r_k) - Q_{\text{max}}) + \beta_{\text{tw}} \sum_{i=1}^N \max(0, t_i - \text{TW}_i^{\text{end}})$$

BI-SUBPOPULATION DIVERSITY RANK

EQUATION (T3-2)

$$\text{BiRank}(S) = \text{rank}_{\text{fit}}(\Phi(S)) + \left(1 - \frac{\text{iter}}{\text{maxIter}}\right) \cdot \text{rank}_{\text{div}}(\Delta(S, \mathcal{P}))$$

• HUMAN-READABLE PLAIN ENGLISH TRANSLATION

Plain English Meaning: The genetic algorithm scores candidate warehouse tours (S) by their total travel distance plus heavy financial penalties for exceeding payload capacity (β_{cap}) or arriving after customer delivery deadlines (β_{tw}). The BiRank formula prevents premature convergence by rewarding solutions that are both cost-effective AND structurally distinct (high Hamming diversity Δ) from other solutions in the gene pool.

Quantum QAOA Routing Formulation (Ising Spin Glass)

COST HAMILTONIAN & DEGREE PENALTIES

EQUATION (T3-3)

$$H_C = \sum_{(i,j) \in \mathcal{E}} c_{ij} \frac{I - Z_i Z_j}{2} + \lambda_{\text{deg}} \sum_{i=1}^N \left(2 - \sum_{j \in \delta(i)} \frac{I - Z_i Z_j}{2}\right)^2$$

• HUMAN-READABLE PLAIN ENGLISH TRANSLATION

Plain English Meaning: Classical binary routing variables ($x_{ij} \in \{0, 1\}$) are mapped to quantum Pauli- Z spin operators ($Z_i \in \{-1, +1\}$). The first term minimizes the road transit cost c_{ij} , while the second term imposes a steep energy penalty if any warehouse stop does not have exactly one inbound and one outbound edge (degree = 2). Ground state bitstrings correspond to valid closed-loop routes.

3.4 Tier 4: Multi-Agent Kinematics & SIPP Trajectory

Tier 4 transforms topological graph tours into continuous time-space kinematic trajectories, ensuring dynamic collision avoidance and industrial safety standard compliance.

CONTINUOUS SWEPT-VOLUME NON-OVERLAP CLEARANCE

EQUATION (T4-1)

$$\min_{t \in [0, T]} \|\mathbf{p}_a(t) - \mathbf{p}_b(t)\|_2 \geq 2R_{\text{swept}} + d_{\text{safety}} \quad \forall a \neq b$$

ISO 3691-4:2023 SAFE KINEMATIC DECELERATION ENVELOPE

EQUATION (T4-2)

$$s_{\text{stop}}(v) = \frac{v^2}{2a_{\text{max}}} + v \cdot t_{\text{reaction}} + s_{\text{margin}}$$

HUMAN-READABLE PLAIN ENGLISH TRANSLATION

Plain English Meaning:
1. Collision Clearance: At every millisecond t , the physical distance between any two AMRs a and b must be greater than twice their robot bounding radius plus safety buffer d_{safety} (0.5 m). Safe Interval Path Planning (SIPP) schedules robot speeds so they never cross intersections simultaneously.
2. ISO 3691-4 Emergency Braking: If an obstruction is detected by onboard LiDAR, the robot must be able to stop completely within distance s_{stop} based on its current velocity v , reaction time (0.1 s), and braking deceleration (1.5 m/s^2).

PARAMETER	STANDARD / FORMULA	ENGINE VALUE	INDUSTRIAL SAFETY ROLE
R_{swept}	Robot Physical Radius	0.45 m	Circular footprint enclosing AMR chassis and overhang cargo.
d_{safety}	Protective Laser Field	0.50 m	LiDAR dynamic protective safety field per ISO 3691-4.
a_{max}	Emergency Deceleration	1.50 m/s^2	Maximum mechanical deceleration without cargo shedding.
t_{reaction}	Safety PLC Response	100 ms	Total latency for sensor detection, bus transmit, and brake bite.

4. API, Telemetry & Swagger Integration Layer

The backend architecture exposes a high-throughput asynchronous REST/WebSocket service conforming to **OpenAPI 3.1.0**, powered by `standalone_server.py` and embedded Swagger UI 5.x.

METHOD	ENDPOINT	RESPONSE DTO	FUNCTIONAL DESCRIPTION
GET	/api/v1/health	HealthDTO	System heartbeats, memory utilization, and active AMR node states.
POST	/api/v1/dispatch/wave	WaveResultDTO	Executes full 4-tier dispatch pipeline with quantum/classical solvers.
GET	/api/v1/dispatch/status/{id}	AuditReportDTO	Queries gate validation proofs and Benders decomposition iterations.
GET	/api/v1/simulation/stream	SSE / FrameDTO	Real-time 60 FPS robot coordinate streaming for digital twin canvas.
GET	/docs	HTML (Swagger UI)	Interactive in-browser testing interface and OpenAPI schema explorer.

5. WMS Quantum Digital Twin 3D Simulator

The frontend is an interactive industrial digital twin built with React 18, Vite, and Three.js, rendering a physical 150 m × 100 m warehouse floor.

FLOOR SURFACE CONTAINMENT GUARANTEE (ZERO-LEAKAGE ASSURANCE)

Every stop $\mathbf{p}(t) = (x, y, z)$ on every AMR tour is strictly bounded within the physical epoxy surface:

$$\forall \mathbf{p}(t) \in \text{Tour} : \quad 0 \leq p_x(t) \leq 150 \text{ m}, \quad 0 \leq p_z(t) \leq 100 \text{ m}, \quad d_{\text{wall}}^{\min} \geq 2.0 \text{ m}$$

Key Digital Twin Modules

- **3D Scene Legend & Active Tour Explorer:** Drawer panel with 4 tabs (3D Objects, Active Tours, Color Guide, Kinematics) providing complete stop-by-stop itineraries (REPLENISH, PICKUP, DROP, DOCK) and "Focus in 3D" tracking cameras.
- **Tiers Optimization Studio:** Interactive deep-dive into Tiers 1-4 with live KaTeX mathematical formulations, parameter sensitivity tuning, and problem solving workflows.
- **Parameter Sweep Studio:** Multi-dimensional sensitivity analysis exploring batch size, quantum qubit limits, and AMR fleet velocity trade-offs.
- **Analytics Studio:** Energy landscape convergence curves, Pareto frontiers, and LIFO extraction DAG visualizers.

6. Technology Stack & Instrument Matrix

LAYER	INSTRUMENT / TECHNOLOGY	VERSION	ARCHITECTURAL FUNCTION
Quantum Optimization	Classiq SDK & Qiskit	0.46+	Quantum circuit synthesis, QMOD modeling, QAOA ansatz generation.
Constraint Programming	Google OR-Tools	9.8+	CP-SAT solver for 3D container bin packing and LIFO DAG ordering.
Metaheuristics	PyVRP / Custom HGS	0.8+	Hybrid Genetic Search with Advanced Diversity Control for VRP.
Backend Framework	Python / FastAPI	3.11+	Asynchronous orchestrator, SIPP kinematic engine, Benders recourse.
Database & Audit	SQLite 3 (WAL Mode)	3.42+	ACID persistence of dispatch waves, audit proofs, and telemetry logs.
Frontend Engine	React 18 & TypeScript	18.2+	Declarative reactive user interface and state synchronization.
3D Graphics	Three.js	0.160+	Hardware-accelerated WebGL 3D warehouse twin visualization.
Math Typography	KaTeX	0.16.9+	High-performance client-side LaTeX formula rendering.
Cloud Deployment	Firebase Hosting	13.x	Global low-latency CDN distribution for digital twin web app.

7. Verification Methodology & Invariant Certification

The system implements a zero-defect deployment methodology where each execution wave is audited across 4 mathematical invariant gates:

- **Gate 1 (Batching Feasibility):** Verifies universal order assignment ($\sum_k u_{ik} = 1$) and vehicle weight limits.
- **Gate 2 (Packing Feasibility):** Verifies pairwise 3D bounding box separation and topological acyclicity of the LIFO DAG.
- **Gate 3 (Routing Feasibility):** Confirms subtour elimination and strict arrival within customer SLA time windows.
- **Gate 4 (Kinematic Feasibility):** Validates continuous swept-volume non-overlap clearance and ISO 3691-4 stopping profiles.

When all 4 gates pass validation, the wave execution is cryptographically stamped with the **Code Verified** certification status.

8. Deployment & Operational Runbook

PRODUCTION OPERATIONAL COMMANDS**1. Launch Standalone API Server:**

```
& 'classiq_env\Scripts\python.exe' -m DispatchEngine.api.standalone_server
```

Swagger UI accessible at: <http://127.0.0.1:8080/docs>

2. Launch Web Simulator Locally:

```
cd web_simulator && npm run dev
```

Digital Twin UI accessible at: <http://localhost:3000/>

3. Deploy Production Simulator to Firebase Hosting:

```
cd web_simulator && npm run build && npx firebase-tools deploy --only hosting
```

Live URL: <https://acoustic-architect-3cgfo.web.app>